Nitrogen Cycle

An important process in ecosystems is the recycling of nitrogen through its living (biotic) and non-living (abiotic) components. The living components, or biota, of the ecosystem participate in the nitrogen cycle in a number of ways.

> If you look closely at the diagram, you will notice that we show the various ways in which nitrogen cycles through nature. **Colour the arrows red (except K) (1 mark)**

Approximately 78% of the air is composed of diatomic nitrogen. Nitrogen is essential to life because it is a key component of amino acids and nucleic acids. Even ATP, the basic energy currency of living things, contains nitrogen. **Colour the air blue**. **(1 mark)**

Neither plants nor animals can obtain nitrogen directly from the atmosphere (A). Instead, they must depend on a process called nitrogen fixation (B). Key players in nitrogen fixation are legumes (C) and the symbiotic bacteria that are associated with their root nodules. Legumes include clover, peas, alfalfa and soy-beans. The bacteria associated with their root nodules are nitrogen-fixing bacteria (D). These bacteria convert nitrogen in the soil to ammonia (NH_3), which can be taken up by some plants. The bacteria and the plants are in a symbiotic relationship. Cyanobacteria are also nitrogen-fixing bacteria: they are prominent in aquatic ecosystems. **Colour parts C and D in green. (1 mark)**

Nitrogen is fixed into the soil through the actions of bacteria and, as we mentioned above, through bacteria that's associated with root nodules of legumes. Both of these methods of fixing nitrogen lead to its incorporation into ammonia (NH_3) in the process known as ammonification (E). The soil is a major reservoir for ammonia and other nitrogen-containing compounds. After nitrogen has been fixed, other bacteria convert it into nitrate, in a process called nitrification (F). In the first step of nitrification, Nitrosomonas (G) convert ammonia to nitrite (NO_2), and in the second step, nitrite is converted to nitrate (NO_3), by Nitrobacter (H). The nitrate (NO_3) is then consumed by plants (I) as the diagram shows. **Colour the nitrogen-fixing bacteria and the nitrogen containing compounds orange. (1 mark)**

However, not all plants consume nitrate: as we mentioned before, some plants are able to use the ammonia from the soil. In both cases, nitrogen enters the primary producers in the biotic community. The plants may then be consumed by animals (J). Herbivores are the primary consumers, and the nitrogen of the plants is used for the synthesis of key organic compounds such as amino acids, proteins and nucleic acids. **Colour a herbivore purple. (1 mark)**

The final aspect of the nitrogen cycle is the process of denitrification (K). This process is performed by a variety of microscopic bacteria, fungi and other organisms. Nitrates in the soil are broken down by these organisms and nitrogen is released into the atmosphere (A). This completes the nitrogen cycle.
Colour the K arrow black. (1 mark)

Questions:

1. What does biotic mean?

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(1 mark)

2. What percentage of the air is nitrogen?

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(1 mark)

3. Why is nitrogen essential for life?

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(1 mark)

4. How do plants get nitrogen if it is not directly from the atmosphere?

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(1 mark)

5. What are nitrogen fixing bacteria?

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(1 mark)

6. What is a major reservoir for ammonia?

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(1 mark)

7. What is a primary consumer?

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(1 mark)

8. Why do herbivores need nitrogen?

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(1 mark)

9. What is denitrification?

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(1 mark)